Conversion of an Entity Relationship Model into a Relational Schema

This assignment aims to convert the provided Entity Relationship (ER) model into a relational database schema. The objective is to design a structured database that facilitates efficient data management for a hotel operating in a tourist area.

Description of the ER Model

The ER model represents the main entities involved in hotel management, including hotels, room types, rooms, categories, and employees. Relationships between these entities are defined using appropriate cardinalities such as one-to-one and one-to-many.

Relational Schema

Based on the ER model, the following relational schema is obtained. Primary keys (PK) and foreign keys (FK) are clearly indicated to ensure data integrity.

HOTEL

HOTEL(
Hotel_id (Primary Key),
Hotel_name,
Type_id (Foreign Key)
)

TYPE

TYPE(
Type_id (Primary Key),
Type_Name
)

ROOM

ROOM(
Room_id (Primary Key),
Floor,
Hotel_id (Foreign Key),
Category_id (Foreign Key)
)

CATEGORY

CATEGORY(
Category_id (Primary Key),
Category_Name,
Price,
Beds_numbers
)

EMPLOYEE

EMPLOYEE(
Employee_id (Primary Key),
Employee_Name,
Employee_Speciality,
Hotel_id (Foreign Key),
Manager_id (Foreign Key referencing EMPLOYEE.Employee_id)
)

Conclusion

This relational schema accurately reflects the structure and constraints of the original Entity Relationship model. It provides a solid foundation for implementing a relational database system that supports efficient hotel data management.